

Bholanath Bala

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RESEARCH INTERESTS

My research interests are in Artificial Intelligence, Deep Learning, and Mathematical Modeling, particularly the use of **transformer-based models** for **signal processing**, **industrial fault diagnosis**, and **intelligent systems**. I am interested in applying models such as **Vision Transformers (ViT)** and **Time-Series Transformers (PatchTST)** to real-world problems and developing reliable and automated solutions for data-driven decision-making.

EDUCATION

Bachelor of Science (B.Sc.) in Electronics and Communication Engineering (ECE).

Khulna University, Bangladesh

CGPA: 3.57/4.00 | Graduated: 2026

- **Undergraduate Thesis:** *A Comparative Study of Vision Transformers and Time-Series Transformers for Bearing Fault Diagnosis Using PU Dataset.*
 - Developed transformer-based deep learning models for automated machinery fault diagnosis.
 - Implemented and evaluated Vision Transformer (ViT) and Time-Series Transformer (PatchTST) architectures.

RESEARCH PUBLICATIONS

Paul, A., **Bala, B.**, Hasan, M. N., & Toma, R. N., "Vision Transformer-Based Classification of Bearing Faults from Vibration Signal Spectrograms," *IEEE-ICCIT* (2025).

- IEEE Xplore: <https://ieeexplore.ieee.org/document/11490241>

Maniruzzaman, M., Salman, R. I., Tawhid, K. G., **Bala, B.**, Hossain, M. M., & Ahsan, M. S., "A Novel Attention-Enhanced Hybrid Deep Learning Model for GI Disease Classification," *IEEE-PECCII* (2026).

- IEEE Xplore: <https://ieeexplore.ieee.org/document/11661820>

ACADEMIC PROJECTS

1. Monitoring Environmental Conditions for Sundari Trees in the Sundarbans.

- Developed an IoT-based environmental monitoring system using sensors to monitor environmental conditions.
- Designed web and mobile interfaces for real-time monitoring and automated alerts.
- Enabled remote access to environmental data for improved monitoring and decision-making.

2. Combating Unpaid Food Purchases Using OpenCV and Face Recognition

- Developed a computer vision system using **Python and OpenCV** to identify unpaid food purchases.
- Applied face recognition techniques to improve automation and security in canteen environments.

3. University Certificate Automation System

- Developed an automated system for managing student certificate approval workflows.
- Designed and implemented a responsive interface using **React.js**.
- Implemented state-driven UI and real-time status tracking.

TECHNICAL SKILLS

- **Programming:** Python, C, Java, JavaScript, TypeScript, MATLAB
- **Web Technologies:** HTML, CSS, Tailwind CSS, React.js
- **Data Science & ML:** NumPy, Pandas, Matplotlib, Machine Learning, Deep Learning
- **Tools:** Git, LaTeX

ACHIEVEMENTS

- **Champion: Research Category, Smart Bangladesh Seminar**, Khulna University, for the Sundarbans Environmental Monitoring Project.
- **2nd Runner-up: Machine Learning Poster Design Competition**, Khulna University.
- **Winner: Open Doors Russian Scholarship Project 2025/2026 (Olympiad)**, Applied Mathematics and Artificial Intelligence.

TRAINING & EXPERIENCE

- **Industrial Training in Data Science at Grameenphone, Bangladesh.**
 - Gained foundational knowledge in data analysis and machine learning workflows through industry-led training sessions

CERTIFICATIONS

- | | |
|---|--|
| • Programming for Everybody (Getting Started with Python), Coursera | • Introduction to Machine Learning, Kaggle |
| • Intermediate Python, DataCamp | • Introduction to SQL, DataCamp |
| | • Introduction to Git, DataCamp |

LANGUAGES

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| • Bangla: Native | • English: Intermediate |
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RECOMMENDATIONS

1. Dr. Rafia Nishat Toma

Associate Professor

Electronics and Communication Engineering (ECE) Discipline

Khulna University, Khulna-9208, Bangladesh

Email: rafiatoma@ece.ku.ac.bd

Relationship: B.Sc. Thesis Supervisor

2. Dr. Md. Maniruzzaman

Professor

Electronics and Communication Engineering (ECE) Discipline

Khulna University, Khulna-9208, Bangladesh

Email: m_m_zaman@ece.ku.ac.bd